



GE Healthcare

Direction 2397208
Revision 3

Signa® 400V/480V TWIN
Catalog M3088TM
Main Disconnect Panel
Service Manual

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Operating Documentation

GE Industrial Systems 55-217305

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A qualified electrician should perform troubleshooting and servicing.



AC and DC control power is present when the breakers are in the open position. De-energize both power sources prior to servicing this panel. If an optional UPS has been added 120V UPS output power is present when the main and branch breakers are open.

This panel has been wired for use at 480V/60HZ, the panel can be converted for use at 380-400-415V/50-60HZ by the changing of the two-transformer primary wiring connections from 480V to 400V. Refer to Section 2-3.

1- THEORY OF OPERATION

This panel incorporates a number of features desirable by MR installations to minimize down time, protect PDU (Power Distribution Unit) electronics, reduce operating costs by maintaining cryogens, and reduce operational delays after any power outage or emergency generator testing. The panel comes wired for a common feed for the PDU, Cryo Cooler MRCC, Gradient MRCC, Cryo Cooler Compressor, and Magnet Monitor with associated auxiliary equipment (MUX BOX, UPS (Uninterruptible Power Supply) and Modem).

PDU branch circuit is controlled by an electrically held contactor, which opens to protect the PDU electronics on any loss of power. The Cryo Cooler MRCC, Gradient MRCC, Cryo Cooler Compressor and UPS input power circuits utilize an adjustable, time delayed, automatic restart control circuit which restores power after any power outage. The time delay auto restart protects the sensitive electronic, control equipment in chillers and compressors from sags and surges which immediately follow power loss from black outs, storms, utility reclosure operations and out of phase automatic transfer switch operations. The 12V DC relay CR1 controls the operation of the Main Power Auto Restarting Contactor, C1. Operation of the 12V DC battery charger and battery provide the control power for relay CR1. Relay CR1 must be energized in order for the Main Power Auto Restarting Contactor to close. The 12V DC auto start control circuit controls all the Cryo Cooler Chiller, Gradient Chiller, Cryo Cooler Compressor, and auxiliary equipment. DC battery control circuit protection time is based upon the condition of the battery but is expected to hold for at least 48 hours. The DC Battery has a life of 5 years at which time the battery and charger must be replaced to assure auto-restart capability.

Power Distribution Unit (PDU), Cryo Cooler MRCC, Gradient MRCC, Cryo Cooler Compressor (CCC), Magnet Monitor, UPS Input Power and UPS output Emergency Off disconnection is also provided by this panel. Two flush-mounted remote emergency off pushbuttons are included with this panel and the cover-mounted emergency off pushbutton provide immediate shutdown for the entire system. Additional remote emergency OFF pushbuttons may be ordered from GE Supply 1-800-200-9760, if required for a particular installation. Remote emergency OFF pushbuttons are listed in Section 5 - Parts List of this manual.

Restoration of power after an emergency off operation is accomplished by first pressing the Main Power ON pushbutton on the cover of the panel and subsequently pressing the PDU ON pushbutton.

The battery charger red indicating light is constantly illuminated while the charging operation is in progress and begins to dim and flicker when charging operation is complete. However, if the red indicating light is OFF either a loss of 120V control power or battery charger failure has occurred.

A circuit breaker provides power for the UPS Input, MUX BOX, Magnet Monitor, and the Modem as indicated on the attached wiring diagram.

1-1 SPECIFICATIONS

1-1-1 ELECTRICAL SPECIFICATIONS

Power: 150A Main Circuit Breaker 380-480V, 350Kcmil maximum conductor size.

- 480V - 90A
- 415V - 103A
- 400V - 107A
- 380V - 109A

*For 480V installations with existing #1 AWG wire main breaker rating plug will be supplied. Contact GE Supply 414-527-6600 if required.

3 Phase, 3 Wire + Ground
480V, 50/60HZ
380, 400 & 415V 50/60HZ

UL 489 & CSA C 22.2

Interrupting Rating 25,000 AIC Symmetrical
50/60 Hertz

IEC 947-2 Interruption Capacity

- 380-415V, U_e , I_{cu} 15,000 RMS
Symmetrical AMPS 50/60 Hertz
- 380-415V, U_e , I_{cs} 10,000 RMS
Symmetrical AMPS 50/60 Hertz

1-1-2 ENVIRONMENTAL SPECIFICATIONS

FOR INDOOR USE ONLY

Temperature 59-89.6F (15-32C)

Humidity 30-75% NON-CONDENSING

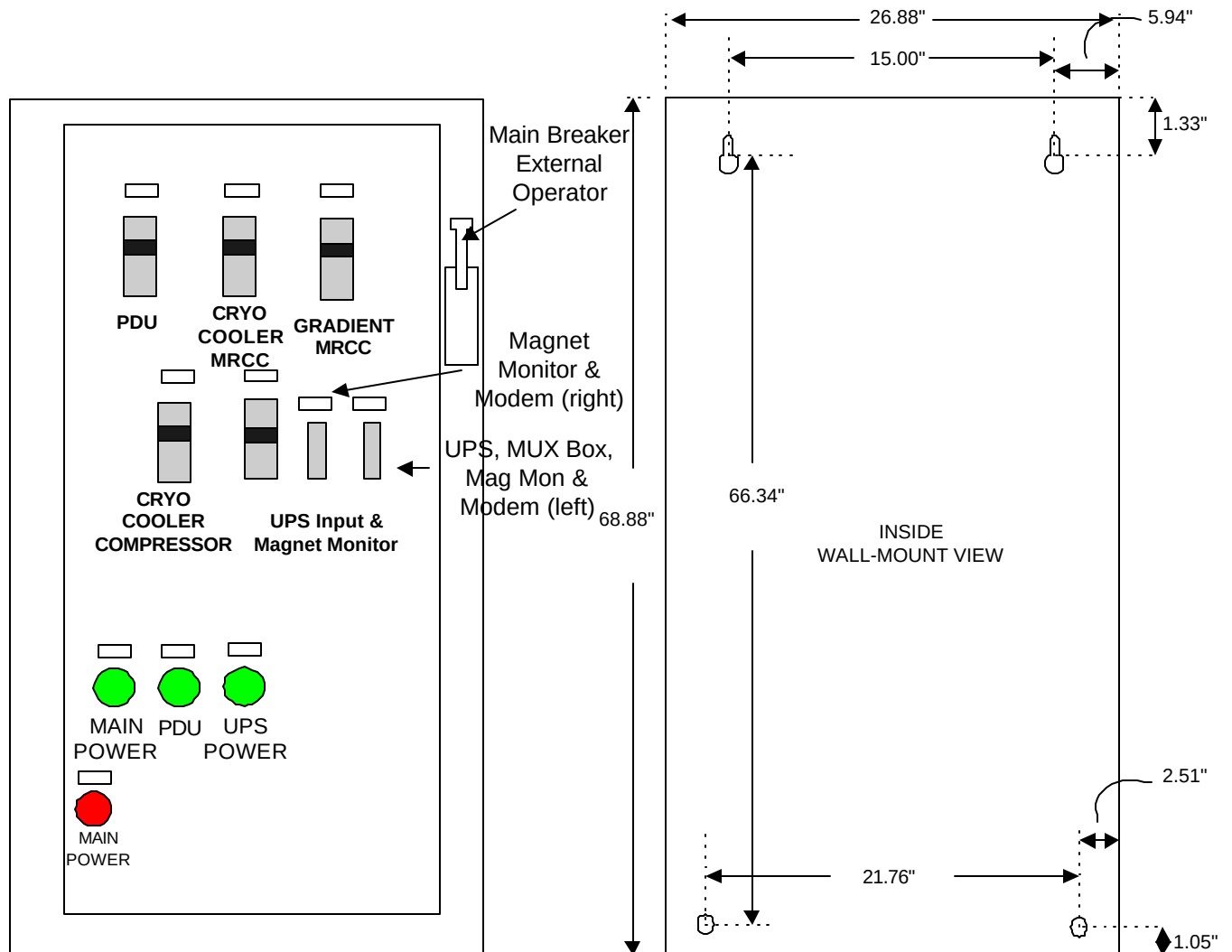
1-2 Panel Dimensions

Refer to Table 1-1 and Illustration 1-1 for the physical dimensions of the panel. The panel may be recessed approximately 7.95 inches (201.9 mm) into the wall for semi-flush installations. See Illustration 1-1 for the front view and mounting pattern of the Main Disconnect Panel.

TABLE 1-1
PANEL EXTERNAL PHYSICAL DIMENSIONS

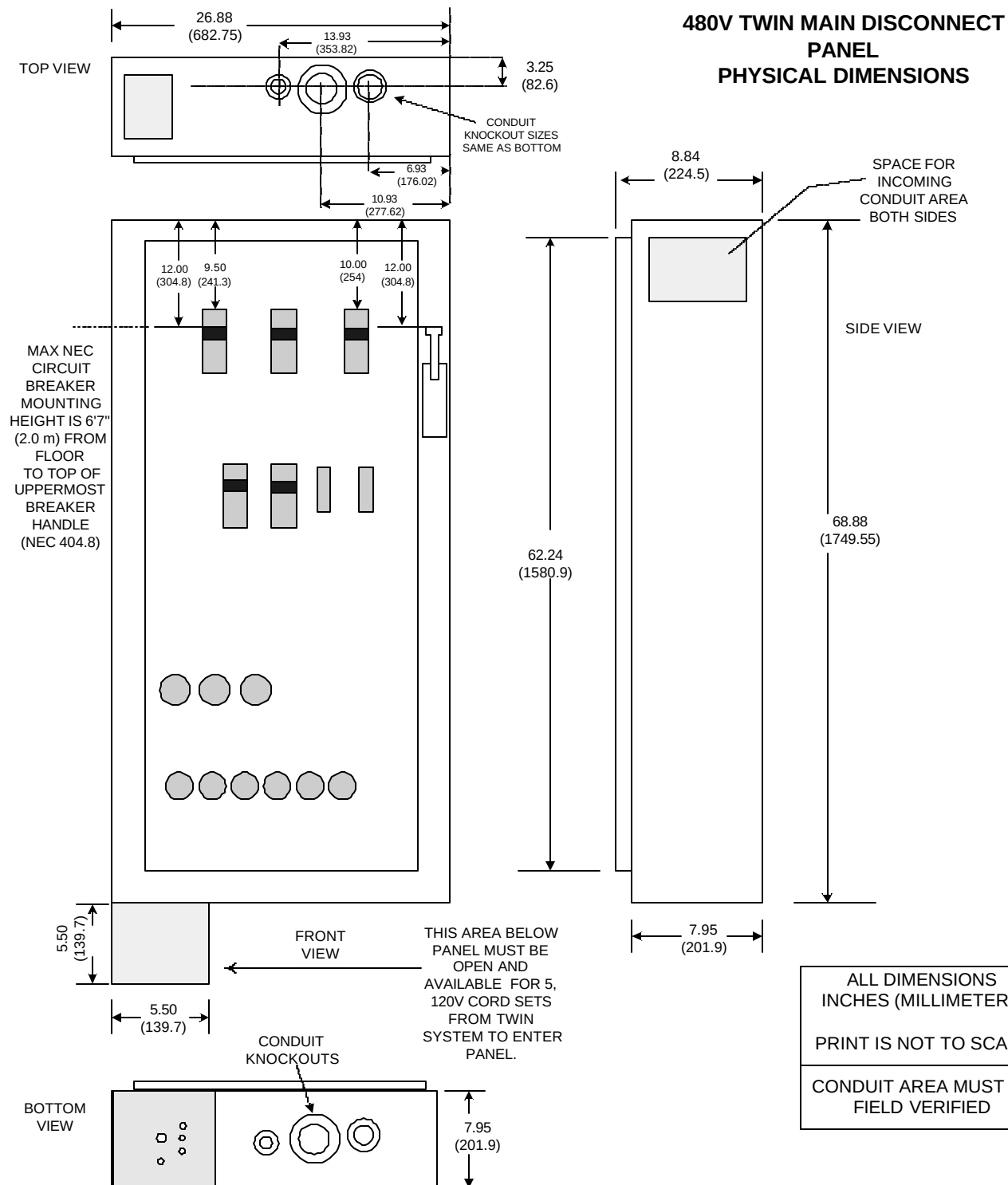
PARAMETER	DIMENSIONS	
Height	68.88 in.	1749.55 mm
Width	26.88 in.	682.75 mm
Depth	8.84 in.	224.5 mm
Weight	350 lbs	158 kg
Door Swing Radius	21.75 in.	552.5 mm

FRONT AND INTERIOR VIEW OF MAIN DISCONNECT PANEL
ILLUSTRATION 1-1



1-3 PANEL LAYOUT AND FULL PHYSICAL DRAWINGS

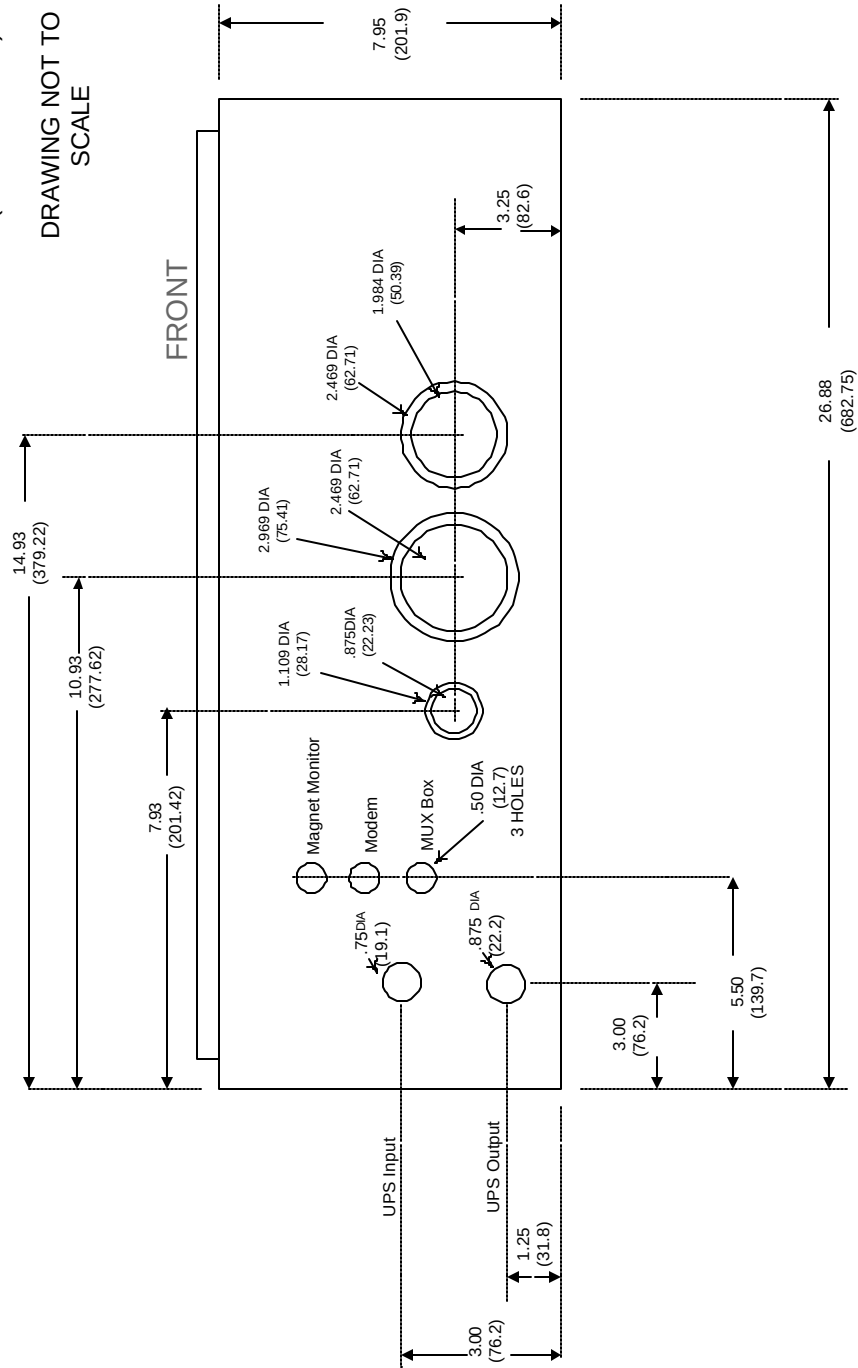
See Illustrations 1-2 and 1-3 for panel layout and full physical drawings including cable access holes and conduit knockout positions.



FULL PHYSICAL DRAWINGS
ILLUSTRATION 1-2

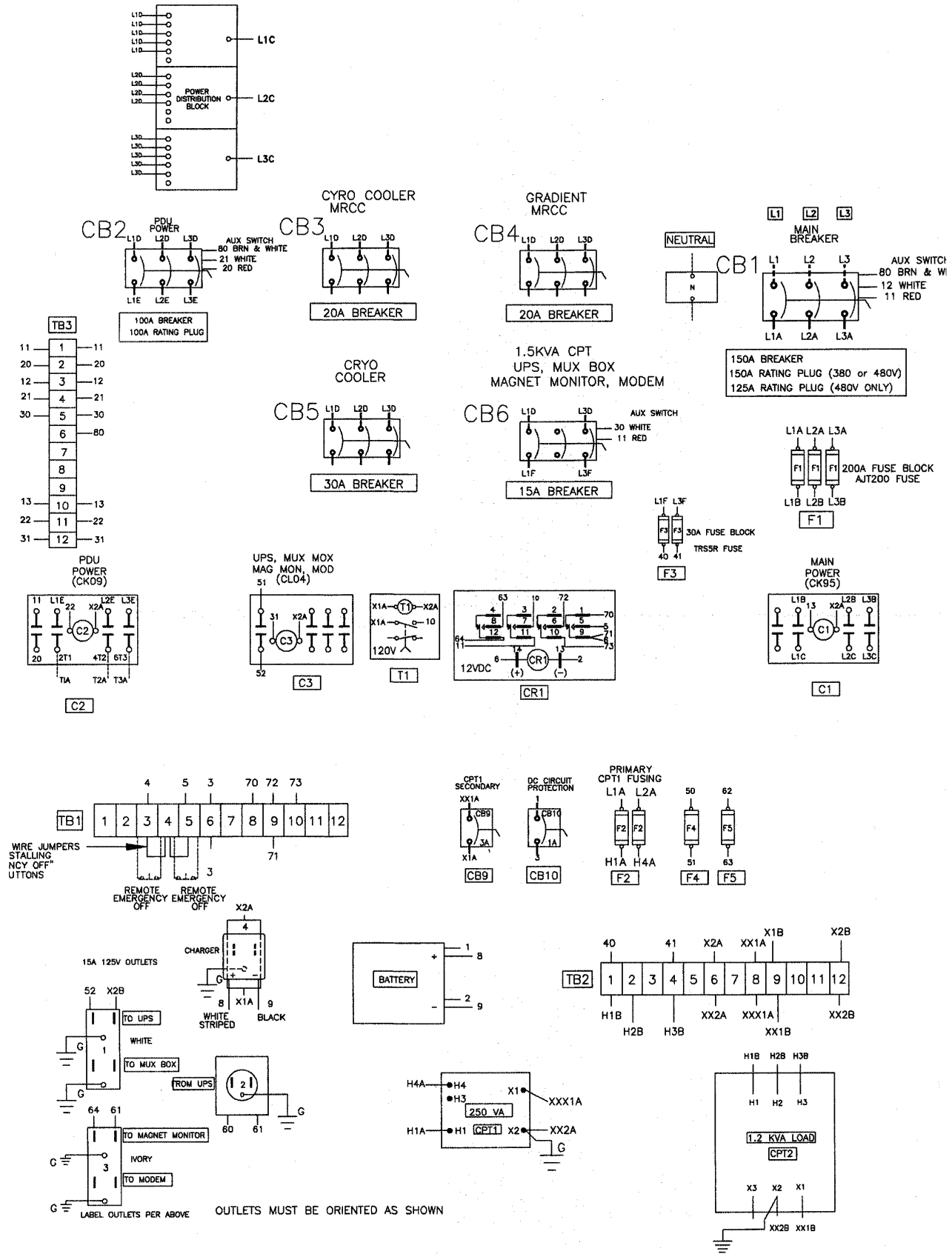
ALL DIMENSIONS
ARE INCHES
(MILLIMETERS)

DRAWING NOT TO
SCALE



DETAILED BOTTOM VIEW
ILLUSTRATION 1-3

400/480V TWIN MAIN DISCONNECT PANEL
DIMENSIONAL DRAWING



INTERNAL LAYOUT OF COMPONENTS
ILLUSTRATION 1-4-1

1-5 Warning Labels

Warning labels shown in this section are provided with the MDP and are to be affixed to the Signa Twin system equipment as indicated.

1-5-1 Cryo Cooler MRCC

Two warning labels are provided with the MDP. Illustration 1-6 shows Cryo Cooler MRCC warning label, which is to be affixed to Cryo Cooler MRCC Compressor Cabinet.

**WARNING: THIS SYSTEM
CRYO COOLER MRCC IS FED BY
AN AUTOMATIC RESTART
CIRCUIT, WHICH ENERGIZES
THE CRYO COOLER CHILLER
AUTOMATICALLY AFTER A
POWER INTERRUPTION.**

CRYO COOLER MRCC LABEL
ILLUSTRATION 1-6

1-5-2 Cryo Cooler Compressor

Two Cryo Cooler Compressor warning labels are provided with the MDP. Illustration 1-7 shows the Cryo Cooler Compressor warning label, which is to be affixed to Cryo Cooler Compressor (2 labels).

**WARNING: THIS CRYO
COOLING COMPRESSOR IS FED
BY AN AUTOMATIC RESTART
CIRCUIT, WHICH ENERGIZES
THE CRYO COOLING
COMPRESSOR
AUTOMATICALLY AFTER A
POWER INTERRUPTION.**

CRYO COOLING COMPRESSOR WARNING LABEL
ILLUSTRATION 1-7

1-5-3 Gradient MRCC Equipment

Two warning labels for the Gradient Chiller are provided with the MDP. Illustration 1-8 shows the Gradient Chiller warning label, which is to be affixed to each of access sides of the Gradient Chiller.

**WARNING: THIS SYSTEM
GRADIENT CHILLER IS FED BY
AN AUTOMATIC RESTART
CIRCUIT, WHICH ENERGIZES
THE GRADIENT CHILLER
AUTOMATICALLY AFTER A
POWER INTERRUPTION.**

GRADIENT CHILLER WARNING LABEL
ILLUSTRATION 1-8

1-5-4 Magnet Monitor Equipment

Four warning labels for the Magnet Monitor equipment are provided with the MDP. Illustration 1-9 shows the Magnet Monitor equipment-warning label, which is to be affixed to each of the following:

- Magnet Monitor box
- UPS for Magnet Monitor
- Modem
- Multiplexer (MUX)

**WARNING: THE MUX
BOX, UPS, MAG MON & MODEM
ARE FED BY AN AUTOMATIC
RESTART CIRCUIT, WHICH
ENERGIZES THESE DEVICES
AUTOMATICALLY AFTER A
POWER INTERRUPTION.**

MUX BOX WARNING LABEL
ILLUSTRATION 1-9

2 INSTALLATION

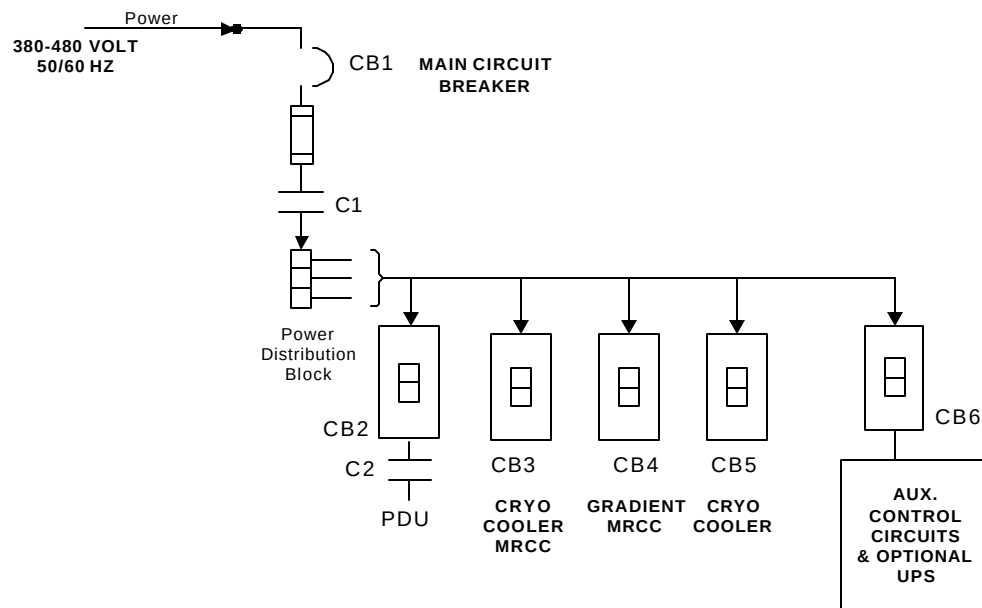
2-1 Installation Single Feed (Refer to Illustration 2-1)

- Incoming power is connected to the main breaker located at the top right side of the Main Disconnect Panel. The preferred location for the incoming conduit is through the top of the enclosure or from the sides near the top. The main breaker maximum wire size is 350kcmil.
- Cryo Cooler MRCC, Cryo Cooler Compressor, and Gradient MRCC are connected directly to the load side of each circuit breaker.
- PDU power is wired to terminals T1A, T2A, and T3A on contactor C2.
- Magnet Monitor Optional UPS power is obtained from this panel from duplex receptacle marked "to UPS" and the UPS output is wired back into this panel to receptacle marked "from UPS".
- Receptacles are provided for the UPS input, UPS output, Magnet Monitor, Modem and MUX BOX. Each receptacle is marked as to function and also designated in panel layout drawing (see Illustration 1-3).
- Access and clear space is required for the UPS input, UPS output, MUX BOX, Modem, and Magnet Monitor cables to enter the bottom left side of the enclosure and for the installation of their plastic strain relief devices. Refer to Illustration 1-2.
- A short, pre-manufactured, heavy-duty extension cord is provided to connect the "to UPS" receptacle to the recessed single receptacle marked "from UPS" for installations without the optional accessory circuit UPS.
- The HD extension cord may be utilized to jumper between "to UPS" and "from UPS" during construction, UPS failure, or omission of UPS.
- Black plastic strain relief devices are shipped with this panel.
- Mounting height of the "top of the grip of the operating handle when in the highest position" of the highest breakers must not exceed 6ft-7in. (2m) above the floor per 2002 National Electric Code (NEC #404.8).
- Inspect and verify that the push on connectors on the battery are connected to the battery terminals and that wires are securely fastened.
- Verify that battery voltage at the terminals of the battery with the charger energized is greater than 12.5VDC and that the battery charger red indicating light is on or flashing.
- Complete the label on the cover of the panel indicating the location and circuit of the power source providing power to this panel.
- The labels that are supplied indicating "Warning of the automatic restart" must be installed on the front cover, and on the rear cover of the Cryo Cooler MRCC. See section 1-5 "Warning Labels".
- The four labels supplied indicating "Warning of automatic restart" must be installed on MUX BOX, UPS, Magnet Monitor, and Modem. See Section 1-5 "Warning Labels".
- Two labels are supplied indicating "Warning of automatic restart". These labels must be installed on the front and rear covers of Cryo Cooler Chiller.
- Two labels are supplied indicating "Warning of automatic restart". These labels must be installed on the front and rear covers of Gradient MRCC.

The control circuit wiring must be completed by installing the two remote emergency off pushbuttons as shown on the wiring diagrams on the inside of the door. The orange labeled, factory installed jumper wire must be removed when installing the remote emergency off push buttons. The Main Contactor or PDU Contactor will not have power unless the remote emergency off pushbuttons are installed between terminals 3, 4, and 5 or temporarily jumpered.

***For warranty parts or technical assistance contact GE SUPPLY – Milwaukee, WI
(414) 527-6600, CST.***

2-2 INCOMING POWER CONFIGURATION



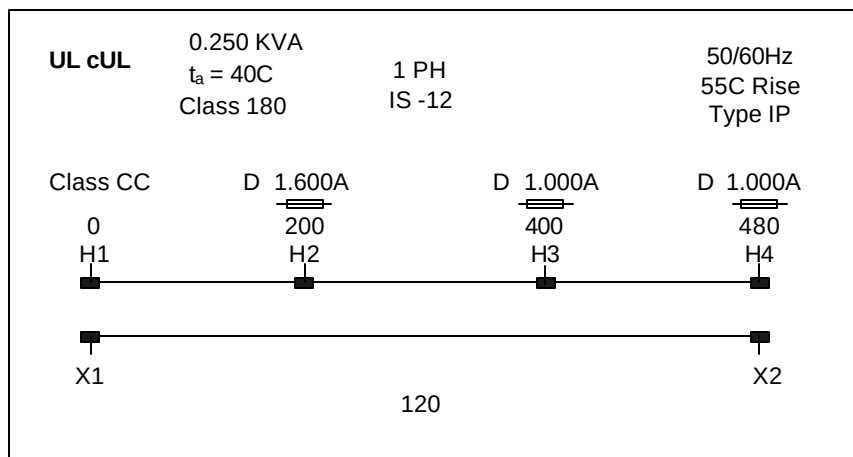
SINGLE FEED SYSTEM CONFIGURATION
ILLUSTRATION 2-1

2-3 480V, 60Hz TO 380-400-415V, 50/60Hz CONVERSION

THIS PANEL HAS BEEN WIRED FOR USE AT 480V/60HZ, THE PANEL CAN BE CONVERTED FOR USE AT 380-400-415V/50-60HZ BY CHANGING THE TWO TRANSFORMER PRIMARY WIRING CONNECTIONS FROM 480V TO 400V. MAKE SURE THE POWER FEED TO THE MDP ARE DISCONNECTED AND LOCKED OFF.

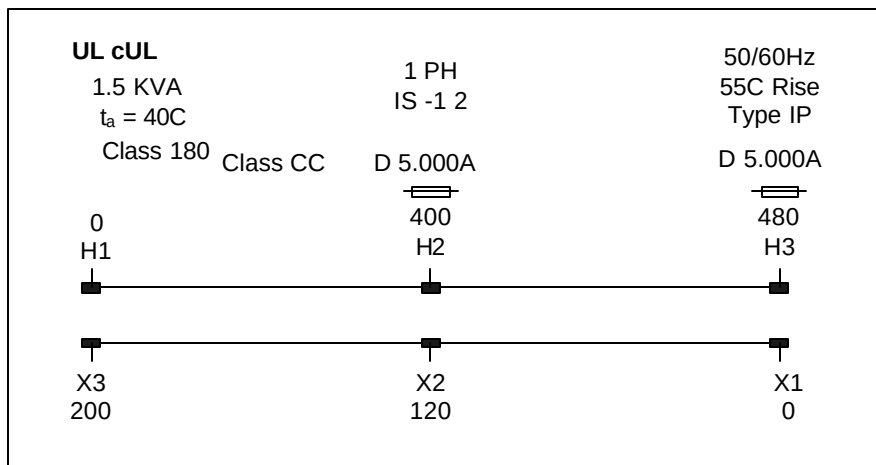
.250KVA Control Transformer, 480V to 380/400/415V Connection

1. Open main circuit breaker and padlock.
2. Open two-pole fuse holder F2.
3. Remove wire originally connected to transformer terminal H4.
4. Connect wire from Step 2 to transformer terminal H3.
5. Close fuse holder F2.
6. Close main circuit breaker.
7. Confirm 120V at CB9 to ground.

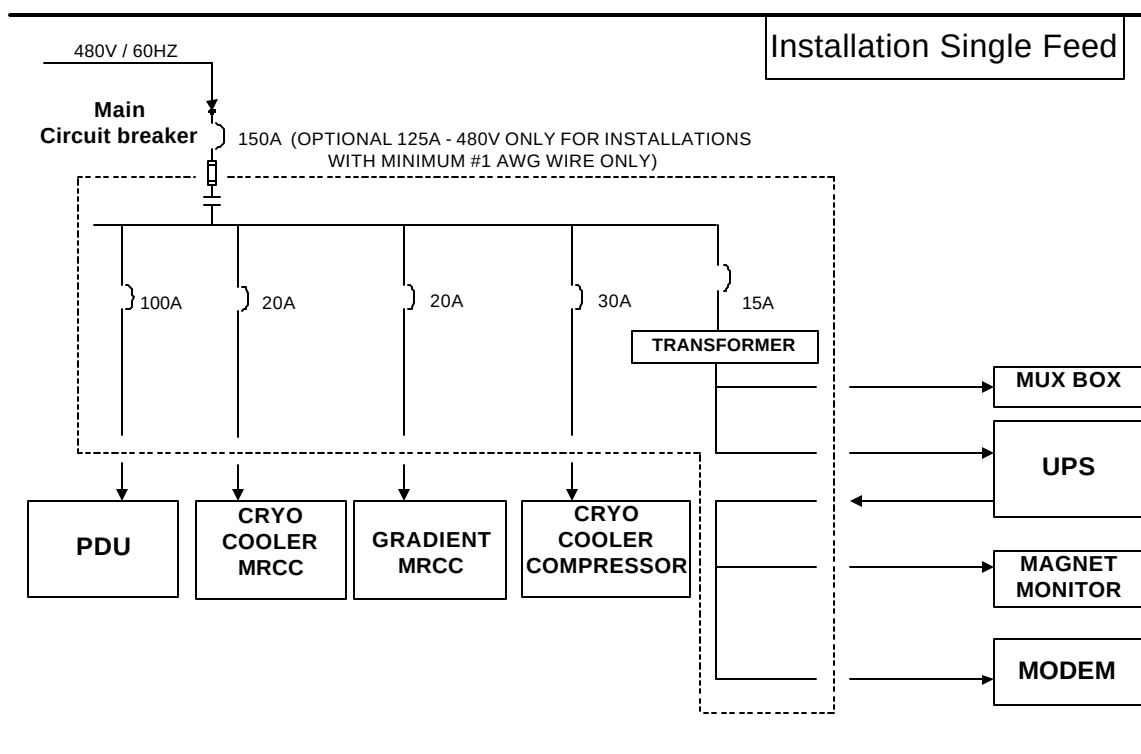


1500VA Transformer, 480V to 380/400/415V Connection

1. Open main circuit breaker and padlock off.
2. Verify that 15-Ampere circuit breaker CB6 is open.
3. Remove two dead front covers for fuse block F3.
4. Remove two TRS5R, 5 ampere fuses.
5. At terminal block TB2 move the wire connected to bp of TB2 terminal 4 to TB2 top of terminal 2.
6. Replace transformer primary fuses in fuse block 3 (FB3) with spare 8A, TRS8R fuses shipped with the panel.
7. Replace the dead front fuse holders on fuse block (FB3).
8. Close main Breaker.
9. Close CB6 and confirm 120V to ground at receptacle marked " To UPS" or CB7.



2-4 POWER CONFIGURATION



INCOMING POWER CONFIGURATION
ILLUSTRATION 2-2

A 480V Optional 125A-rating plug may be used if minimum #1 wire is used. Contact GE Supply 414-527-6600, for exchange of the main circuit breaker rating plug for new catalog # SRPE150A125.

2-5 TESTING SINGLE FEED PANELS

WARNING!

AC AND DC CONTROL POWER IS PRESENT WHEN THE BREAKERS ARE IN THE OPEN POSITION. MAKE SURE NO ONE IS WORKING ON THE EQUIPMENT, WHICH THIS PANEL FEEDS PRIOR TO TESTING THE PANEL.

TESTING ACTION

1. Press the main power ON pushbutton and PDU ON pushbutton. Verify Main Power Contactor; Magnet Monitor and PDU selector switches are all set to ON.
2. Press any emergency off pushbutton.
3. Press the main power ON pushbutton and PDU ON pushbutton. Verify Main Power ON, Magnet Monitor, and PDU selector switches are all set to ON and repeat for the all other emergency off pushbuttons.
4. Press the main power ON pushbutton and PDU ON pushbutton. Verify Main Power ON; Magnet Monitor and PDU selector switches are all set to ON.
5. Press any emergency off pushbutton.
6. Starting with the de-energized state of #5, turn OFF incoming power using the externally operated Main Breaker located at the top right side of this panel.

VERIFY

1. All three green pilot lights should be on indicating proper contactor operation.
2. Both Main Contactor and PDU Contactor must de-energize and stay de-energized. All three green pilot lights must be OFF. Verify that the UPS output is disconnected from Magnet Monitor and Modem.
3. Both Main Contactor and PDU Contactor must de-energize and stay de-energized after an emergency OFF operation. All pilot lights must be OFF when any emergency off is pressed. Verify that the UPS output is disconnected from Magnet Monitor and Modem.
4. All three green pilot lights should be ON indicating proper contactor operation.
5. All contactors must de-energize and stay de-energized. All three green pilot lights must be OFF.
6. All three lights must be OFF indicating loss of power. Confirm disconnection of UPS output from Magnet Monitor and Modem. UPS 120V output power is present at CB8 to ground but disconnected from the Magnet Monitor and Modem by confirming at the Magnet Monitor receptacle that 120 volts is not present.

TESTING ACTION

7. With incoming power OFF from step 6, now restore incoming power to panel (switch main breaker to ON).
8. Press the main power ON pushbutton and PDU pushbutton. Verify Main Power ON; Magnet Monitor and PDU selector switches are all set to ON.
9. Individually rotate on-off selector switch for the main power contactor, and Magnet Monitor OFF then ON.
10. Individually rotate the PDU ON-OFF selector switch for the PDU contactor.
11. Press the main power ON pushbutton and PDU ON pushbutton. Verify Main Power ON; Magnet Monitor and PDU selector switches are all set to ON. Turn OFF incoming power using the panel main circuit breaker of panel and restore power to the panel. This is a simple OFF-then-ON function.

VERIFY

7. Upon restoration of power all contactors must remain OFF. All pilot lights must be OFF. UPS output remains disconnected. This demonstrates emergency off function remains disabled during a loss of power. No auto restart after any emergency OFF operation.
8. All three green pilot lights should be ON.
9. Each green indicating light should turn OFF and the respective contactor should open in OFF. Each green pilot light should turn ON and its respective contactor close when selector switch is switched to ON.
10. The green light for the PDU should turn OFF when the switch is moved to OFF. Moving the switch to ON will not restore the PDU power unless the PDU power ON pushbutton is pressed after the selector switch has been set to ON.
11. All three green indicating lights should turn OFF with loss of power. The Main Power Contactor and Magnet Monitor auxiliary equipment contactor and two green indicating lights must turn ON with the restoration of power indicating the proper operation of the automatic restart feature. PDU pilot light turns OFF on any loss of power and the PDU green pilot light and contactor does not restart automatically.

2-6 FINAL CHECK

1. Verify that charger indicating light is lit or flickering. Confirm battery voltage is great than 12.5VDC.
2. Confirm all wire connections are properly tightened.

3-TROUBLESHOOTING

Auto Restart Function Main Contactor, PDU Contactor, and Magnet Monitor auxiliary equipment will not be de-energized unless the control circuit emergency off pushbuttons are wired as indicated on the wiring diagram. The orange labeled, factory installed jumper wire must be removed when connecting the remote emergency off pushbuttons.

Make sure the emergency off pushbuttons are wired through normally closed contacts.

Operation of any emergency off pushbutton should de-energize relay CR1, shutting the entire system down and disconnecting the UPS output from the Magnet Monitor and Modem.

Check LED lamps for burnt out or broken in shipment.

Check that battery charger is firmly installed in its receptacle. 12V DC output voltage must be 12 to 13.65V DC under normal operating conditions. If DC voltage is less than 11V, replace the charger and battery. 12V DC relay CR1 controls the operation of all branch circuit contactors. Operation of the 12V DC battery charger and battery provide the control power for relay CR1. Relay CR1 must be energized in order for any branch circuit contactor to close.

NOTES:

**FOR WARRANTY PARTS OR TECHNICAL ASSISTANCE CONTACT:
GE SUPPLY – MILWAUKEE, WI. (414)-527-6600 CST**

4-MAINTENANCE

4-1 General Maintenance

Note

The auto restart circuit requires the 7 Ah DC battery to be functional. Semi-annually verify that the 12V DC charging circuit is operational.

Battery and Charger Test

1. Unplug battery charger from receptacle.
2. Remove the (+) wire from the charger.
3. Plug in charger and measure DC voltage of charger.
DC voltage must be between 12 and 13.65 volts. If less than 11 volts, replace charger.
4. With charging circuit de-energized, measure the DC battery voltage. If battery voltage is less than 11 volts replace battery. Whenever a battery is replaced the charger must also be replaced.

Normal operation of DC circuit is from DC charger with battery fully charged in float stage. If battery is suspect or questionable, replace immediately as this is an insignificant maintenance cost. The manufacturer of the battery is B & B Battery and the manufacturer for the charger is EDS, Inc. Contact GE Supply or BatteryStuff.com for battery replacement.

Battery# BP7-12
Charger# 11027

Specific battery maintenance and recharge/discharge information can be found in attached B & B documents and on the B & B web site (below).

Battery life is 5 years at which time it must be replaced. Initial date code on top of battery is read as (mm/dd/y/c). Where c = Q.C. inspector code.

B & B Battery can be located at: www.batterystuff.com

GE Supply
12221 W. Feerick St.
Milwaukee, WI 53222
Phone 414-527-6600
Fax 414-527-6652

GE Supply
National Service Center
9100 Purdue Road Suite 300
Indianapolis, IN 46268
Phone 1-800-243-7313

See illustration 4-1 and 4-2 for battery manufacturer's main data sheet.



Maintenance-Free Rechargeable Sealed Lead-Acid Battery



BP7-12

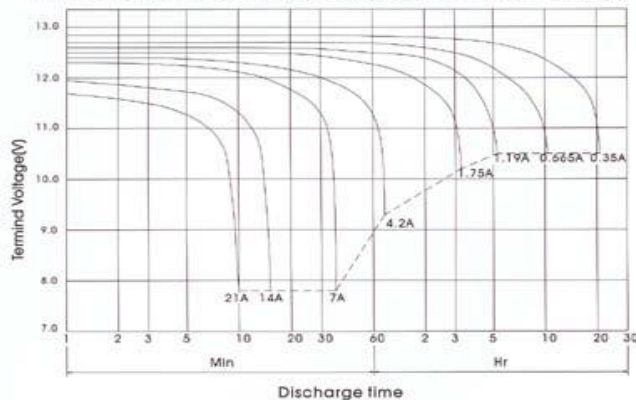
The battery is constructed by plates, separators, safety valves and container. Since the electrolyte is held by a glass-mat separator and plates, the battery can use in any direction and position without leakage.

PERFORMANCE SPECIFICATIONS

Nominal Voltage(V).....	12 volts(6cells in series)
Nominal Capacity(AH)	
20 hour rate F.V.(1.75V/cell) (350mA to 10.50volts)	7.0A.H.
10 hour rate F.V.(1.75V/cell) (665mA to 10.50volts)	6.65A.H.
5 hour rate F.V.(1.75V/cell) (1190mA to 10.50volts)	5.95A.H.
1 hour rate F.V.(1.55V/cell) (4200mA to 9.30volts).....	4.2A.H.
Approximate Weight.....	2600g.(5.73lbs.)
Terminal	
Standard.....	Type T2
Optional.....	Type T1
Internal Resistance (Fully Charged Battery).....	<25m Ω
Maximum Discharge Current For 5 sec.(A).....	105A
Maximum Charge Current(A).....	2.1A
Ambient Temperature	
Charge.....	0°C(32°F)~40°C(104°F)
Discharge.....	-20°C(-4°F)~50°C(122°F)
Storage.....	-20°C(-4°F)~40°C(104°F)
Vibration test:	
Frequency: 16.7HZ	
Amplitude: 4mm	
Vibrate the battery horizontally or vertically for 60 minutes.The battery have no abnormality.	
Case.....	ABS
Dimension(mm/inch)	
Length ± 1.5 mm.....	151/5.95
Width ± 1.5 mm.....	65/2.56
Container Height ± 1.5 mm.....	93/3.66
Total Height ± 2 mm.....	98/3.86
Application.....	UPS.Laboratory Equipment,Toy-Cars,Power Packs,Fishing Lights.

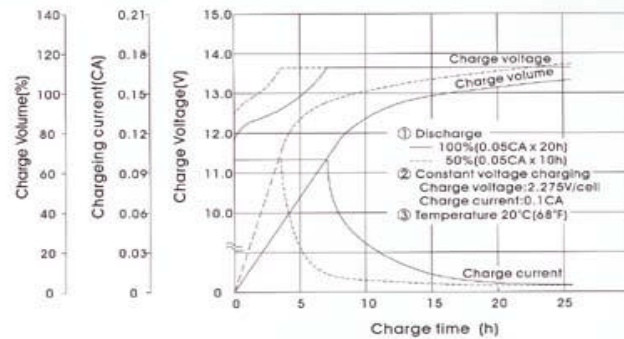
BATTERY MANUFACTURER DATA SHEET
ILLUSTRATION 4-1

BP7-12 Battery discharge characteristics (25°C/77°F)



BATTERY CHARGING CHARACTERISTICS

(Typical example of charge characteristics for the standby use)

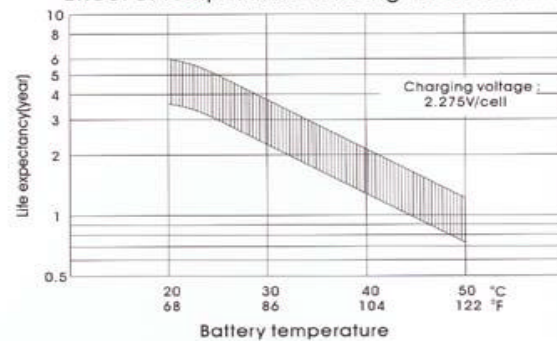


Charging Procedure

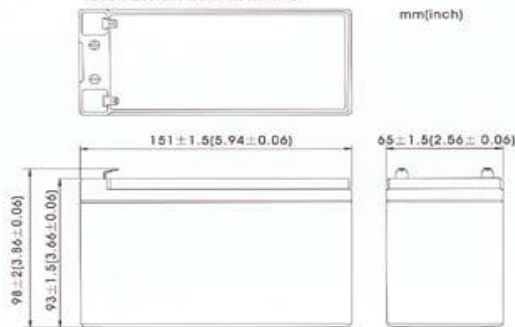
Application	Charging method	Charging Voltage at 20°C (V/cell)	Temperature compensation coefficient of charging voltage (mV/°C/cell)	Max. charging current (CA)	Charging time 0.1CA, 20°C (h)		Temp (°C)
					100% discharge	50% discharge	
For standby power source	Constant voltage & Constant current charging	2.25-2.30	-3.3 (-1.8mV/°C/cell)	0.3	24	20	0-40 (32-104°F)
For cycle service	Constant current charging (with current restriction)	2.40-2.50	-5 (-2.8mV/°C/cell)	0.3	16	10	

*Temperature compensation of charging voltage is not needed when using the batteries within 5°C to 35°C range.

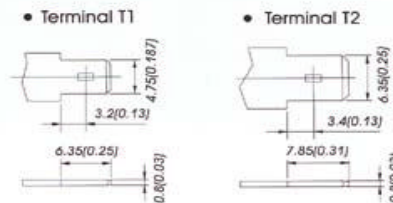
Effect of Temperature on Long Term Float Life



OUTER DIMENSIONS



TERMINAL TYPE



TERMINAL POSITION



B.B. BATTERY CO., LTD.

USA :
B&B BATTERY (USA) INC.
6415 RANDOLPH ST. CITY OF
COMMERCE, C.A. 90040 U.S.A.
TEL: 1-323-278-1900
FAX: 1-323-278-1268 1-800-278-8599
Web Site: <http://www.bb-battery.com>

CHINA FACTORY:
CHENG DONG TRIAL AREA,
HUANG GANG, RAOPING
GUANG DONG, CHINA, 515700
TEL: 86-768-7601001-2
FAX: 86-768-7601469
E-mail: bbchina@pub.chaozhou.gd.cn

UK:
B & B BATTERY (UK) CO. LTD.
32 MOOR FIELD
WHALLEY, CLITHEROE
LANCS, ENGLAND BB7 9SA
TEL: 44-(0)1254-824726
FAX: 44-(0)1254-823228

HONG KONG OFFICE:
NATIONAL TRADING LTD.
6D HILTON TOWER, 96
GRANVILLE RD, TSIMSHATSUI
EAST, KLN, HONG KONG.
TEL: 852-2301-3800
FAX: 852-2739-1182
E-mail: bbhk@hkstar.com

JAPAN:
B & B BATTERY (JAPAN) CO. LTD.
1375-11 NARAHARA-MACHI
HACHIOJI, TOKYO 193-0803, JAPAN
TEL: 81-426-256-537
FAX: 81-426-256-537

TAIWAN OFFICE:
B & B BATTERY CO. LTD.
TEL: 886-2-22990887 22990850
FAX: 886-2-22989702



MH19884



BATTERY MANUFACTURER DATA SHEET ILLUSTRATION 4-2

4-2 Battery Manufacturer's Application Notes

BB BP batteries are highly efficient maintenance free electrochemical systems designed to provide years of trouble free electrical energy. The performance and service life of these batteries can be maximized by observing the following guidelines:

1. Heat kills batteries. Avoid placing batteries in close proximity to heat sources of any kind. The longest service life will be attained where the battery temperature does not exceed 20°C. (also see notes 3 & 8 hereunder) . When calculating the correct float voltage setting, whether or not temperature compensation is required, full consideration must be given to the temperature of the battery and room ambient. For the purpose of the calculation, consider the temperature of a battery on float to be 1°C. above local ambient. Also, if the battery is used in an enclosure, the temperature gradient of the enclosure itself must be included in the calculation. i.e. The operating temperature of the battery is given by: -Room temperature + enclosure temperature +1°C.
2. Since a battery may generate ignitable gases, do not install close to any equipment that can produce electrical discharges in the form of sparks.
3. When the battery is operated in a confined space, adequate ventilation should be provided.
4. The battery case is manufactured from high impact ABS plastic resin. It should not be placed in an atmosphere of, or in contact with organic solvents or adhesive materials.
5. Correct terminals should be used on battery connecting wires. Soldering is not recommended but if unavoidable please refer to us for further guidance.
6. Avoid operating at temperatures outside the range -15 to +50°C. for float /standby applications and +5 to +35°C. for cyclic use.
7. When there is a possibility of the battery being subjected to heavy vibration or mechanical shock, it should be fastened securely and the use of shock absorbent material is advisable.
8. When connecting the batteries, free air space must be provided between each battery. The recommended minimum space between batteries is 0.02 inches (5mm) to 0.04 inches (10mm) . In all installations due consideration must be given to adequate ventilation for the purposes of cooling.
9. When the batteries are to be assembled in series to provide more than 100V, proper handling and safety procedures must be observed to prevent accidental electric shock. (See note #15 below) .
10. If 2 or more battery groups are to be used, connected in parallel, they must be connected to the load through lengths of wires, cables or busbars that have the same loop line resistance as each other. This makes sure that each parallel bank of batteries presents the same impedance to the load as any other of the parallel banks thereby ensuring correct equalization of the source to allow for maximum energy transfer to the load.
11. To obtain maximum life, the ripple current flowing in the battery, from any source, should not exceed 0.1C Amps R.M.S.
12. When cleaning the battery case, ALWAYS use a water soaked wet or dampened cloth but NEVER use oils, organic solvents such as petrol, paint thinners etc. DO NOT even use a cloth that is impregnated or has been in contact with any of these or similar substances.
13. Do not attempt to dismantle the battery. If accidental skin /eye contact is made with the electrolyte, wash or bathe the affected area /part straight away with liberal amounts of clean fresh water and seek IMMEDIATE medical attention.
14. DO NOT INCINERATE batteries as they are liable to rupture if placed into a fire. Batteries, that have reached the end of their service life, can be returned to us for safe disposal.
15. Touching electrically conductive parts might result in an electric shock. Be sure to wear rubber gloves before inspection or maintenance work.
16. The use of mixed batteries with different capacities, that may have been subjected to different uses, be of different ages and are of different manufacturers is liable to cause damage to the battery itself and /or the associated equipment. If this is unavoidable please consult us beforehand.
17. To obtain maximum life, batteries should never be stored in a discharged state.
18. In order to obtain maximum working life, when the batteries are used in an UPS system, the following is advised:-
 - (a) Where the D.C. Input exceeds 60 volts, each battery should be insulated from the battery stand by using suitable polypropylene or polyethylene material.
 - (b) In high voltage systems the resistance between battery and stand should always be greater than 1 Megohm. An appropriate alarm circuit could be incorporated to monitor any current flow.

4-3 Battery Manufacturer's Material Safety Data Sheet

B & B Battery (U.S.A.), Inc.

6414 Randolph Street, City of Commerce, CA 90040 (323) 278-1900 Fax (323) 278-1268

MATERIAL SAFETY DATA SHEET

PRODUCT NAME: Sealed Maintenance free Lead Acid Batteries

DATE: 7/1/00

ISSUED BY: ENGINEERING

TELEPHONE NO: (323) 278-1900

HAZARDOUS COMPONENTS

COMPONENTS	%WEIGHT	TLV	LD50 ORAL	LC50 INHALATION	LC50 CONTACT
Lead (Pb, PbO ₂ , PbSo)	About 70%	N/A	(500) Mg/Kg	N/A	N/A
Sulfuric Acid	About 20%	1 mg/m ³	(2.140) Mg/Kg	N/A	N/A
Fiberglass Separator	About 5%	N/A	N/A	N/A	N/A
Styron R 478 (Polystyrene)	About 5%	N/A	N/A	N/A	N/A

PHYSICAL DATA

COMPONENTS	DENSITY	MELTING POINT	SOLLUBILITY (H ₂ O)	ODOR	APPEARANCE
Lead	11.34	327.4°C (Boiling)	None	None	Sliver-Gray Metal
Lead Sulfate	6.2	1070°C (Boiling)	40 mg/l (15°C)	None	White Powder
Lead Dioxide	9.4	290°C (Boiling)	None	None	Brown Powder
Sulfuric Acid	About 1.3	About 114°C (Boiling)	100%	Acidic	Clear Colorless Liquid
Fiberglass Sep.	N/A	N/A	SLIGHT	TOXIC	WHITE FIBROUS GLASS
478 Polystyrene	N/A	N/A	NONE	NO ODOR	SOLID

FLAMMABILITY DATA

COMPONENTS	FLASHPOINT	EXPLOSIVE LIMITS	COMMENTS
Lead	None	None	
Sulfuric Acid	None	None	
Hydrogen		4% - 74.2%	Sealed batteries can emit hydrogen only if over charged (Float voltage > 2.4 VPC)
Fiberglass Sep.	N/A	N/A	Toxic vapors may be released. In case of fire: wear self-contained breathing apparatus.
478 Polystyrene	None	N/A	Temperatures over 300 °C (572°F) may release combustible gases. In case of fire: wear positive pressure self-contained breathing apparatus.

FIRST AID

SULFURIC ACID PRECAUTIONS

SKIN CONTACT: Flush with water, see physician if contact area is large or if blisters form.

EYE CONTACT: Call physician immediately and flush with water until physician arrives.

Ingestion: Call physician. If patient is conscious, flush mouth with water, have the patient drink milk or sodium bicarbonate solution.

DO NOT GIVE ANYTHING TO AN UNCONSCIOUS PERSON.

**B & B MATERIAL SAFETY DATA SHEET FOR BP7-12 BATTERY
ILLUSTRATION 4-3**

REACTIVITY DATA

COMPONENT	Sulfuric Acid
STABILITY	Stable at all temperatures
POLYMERIZATION	Will not polymerize
INCOMPATIBILITY	Reactive metals, strong bases, most organic compounds
DECOMPOSITION PRODUCTS	Sulfuric dioxide, trioxide, hydrogen sulfide, hydrogen
CONDITIONS TO AVOID	Prohibit smoking, sparks, etc. from battery charging area. Avoid mixing acid with other chemicals.

SPILL OR LEAK PROCEDURES

STEPS TO TAKE IN CASE OF LEAKS OR SPILLS

If sulfuric acid is spilled from a battery, neutralize the acid with sodium bicarbonate (baking soda), sodium carbon (soda ash), or calcium oxide (lime).

Flush the area with water discard to the sewage systems. Do not allow unneutralized acid into the sewage system.

WASTE DISPOSAL METHOD:

Neutralized acid may be flushed down the sewer. Spent batteries must be treated as hazardous waste and disposed of according to local state, and federal regulations. A copy of this material safety data must be supplied to any scrap dealer or secondary smelter with battery.

PROTECTION

EXPOSURE	PROTECTION	COMMENTS
SKIN	Rubber gloves, Apron	Protective equipment must be worn if battery is cracked or otherwise damaged.
RESPIRATORY	Respirator (for lead)	A respirator should be worn during reclaim operations if the TLV exceeded.
EYES	Safety goggles, Face Shield	

ELECTRICAL SAFETY

Due to the battery's low internal resistance and high power density. High levels of short circuit can be developed across the battery terminals. Do not rest tools or cables on the battery. Use insulated tools only.

Follow all installation instruction and diagrams when installing or maintaining battery systems.

HEALTH HAZARD DATA

LEAD: The toxic effects of lead are accumulative and slow to appear. It affects the kidneys, reproductive, and central nervous system.

The symptoms of lead overexposure are anemia, vomiting, headache, stomach pain (lead colic), dizziness, loss of appetite, and muscle and joint pain. Exposure to lead from a battery most often occurs during lead reclaim operations through the breathing or ingestion of lead dusts and fumes.

THIS DATA MUST BE PASSED TO ANY SCRAP OR SMELTER WHEN A BATTERY IS RESOLD.

SULFURIC ACID: Sulfuric acid is a strong corrosive. Contact with acid can cause severe burns on the skin and in the eyes. Ingestion of sulfuric acid will cause GI tract burns. Acid can be release if the battery case is damaged or if the vents are tampered with.

FIBERGLASS SEPARATOR: Fibrous glass is an irritant of the upper respiratory tract, skin and eyes. For exposure up to 10F/CC use MSA Comfoll with type H filter. Above 10F/CC up to 50F/CC use Ultra -Twin with type H filter.

This product is not considered carcinogenic by NTP or OSHA.

4-4 Battery Features

The following is a list of features of the B & B batteries.

SEALED CONSTRUCTION

BB unique construction and sealing technique ensures that no electrolyte leakage should occur from the terminals or case of any BP battery. This feature provides for safe and efficient operation of BP batteries in any orientation. BB BP batteries are classified as “Non-Spill able” and meet all requirements of the International Air Transport Association. (I.A.T.A. Dangerous Goods Regulations).

ELECTROLYTE SUSPENSION SYSTEM

All BB BP batteries utilize an electrolyte suspension system consisting of a glass fiber separator material. This suspension system helps to achieve maximum service life, by fully retaining the electrolyte and preventing its escape from the separator material. No silica gels or other contaminants are used.

GAS GENERATION

BP batteries incorporate a unique BB design that effectively recombines over 99% of the gas generated during normal usage.

MAINTENANCE FREE OPERATION

During the life of BP batteries, there is no need to check their specific gravity or add water etc. In fact, there are no provisions for such maintenance functions to be carried out.

OPERATION IN ANY ORIENTATION

The combination of sealed construction and BB’s electrolyte suspension system permits operation of BP batteries in any orientation without loss of capacity, electrolyte, or service life.

LOW PRESSURE VENTING SYSTEM

BB BP batteries are equipped with a safe, low pressure venting system, which is designed to release excess gas and reseal automatically in the event of the internal gas pressure rising to an unacceptable level. This low pressure venting system, coupled with the significantly high recombination efficiency, make BB BP batteries one of the safest valve regulated lead acid batteries available.

HEAVY DUTY GRIDS

The heavy-duty lead calcium alloy grids in BP batteries provide an extra margin of performance and service life in both float and cyclic applications, even in conditions of deep discharge.

CYCLIC SERVICE LIFE

Depending upon the average depth of discharge, over 1,000 discharge/recharge cycles can be expected from BP batteries.

FLOAT SERVICE LIFE

The expected service life of BP batteries used in standby applications is typically 5 years; however, experience has shown that their service life often exceeds 6 years, if the BP batteries are operated strictly within specification.

4-5 General Battery and Charger Specifications

Refer to Tables 4-1 and 4-2 for manufacturer's general specifications for the B & B BP7-12 battery and the EDS 11027 charger.

TABLE 4-1
GENERAL BATTERY SPECIFICATIONS

B&B Model #	Nominal Voltage (V)	Nominal Capacity 20 hr rate (mA)	Discharge Current 20 hr rate (mA)	Dimensions								Approx. Weight		Standard Terminal Type
				Length		Width		Height		H.O.T.				
				in.	mm.	in.	mm.	in.	mm.	in.	mm.	lbs.	kg.	
BP7-12	12	7.0	350	5.95	151	2.56	65	3.7	94	3.86	98	5.7	2.6	T2

GENERAL CHARGER SPECIFICATIONS

TABLE 4-2

EDS Model#	Nom. Voltage (V)	Voltage Range (V)	Nominal Current (A)	Type of Charger
11027	12	13.6- 14.70	0.70	Dual Voltage Auto

Charger Dimensions							
Length		Width		Height		Approx. Weight	
Inches	mm	inches	mm	inches	mm	lbs	kg
3.4	86.4	2.8	71.1	2.3	58.42	1.3	0.59

5-PARTS LIST

5-1 Renewable Parts

See Table 5-1 for renewable parts for the Twin System main disconnect panel.

Table 5-1
LIST FOR RENEWABLE PARTS

			Manufacturer Part Number		
Part	Volts	Amps	Buss	Gould-Shawmut	GEMS Cat #
Fuse F1	480	200	LPJ200	AJT200	2375862-2
Fuse F2	480	1.25	SNQ-R-1 1/4	ATQR1 1/4	2276547-2
Fuse F3	480	5	FRS-R-5	TRS5R	2276547-21
	400	8	FRS-R-8	TRS8R	2276547-6
Fuse F4	120	10	FNQ-R-10	ATQR10	2276547-11
Fuse F5	120	10	FNQ-R-10	ATQR10	2276547-11
Red PB Operator*			CR104PBG00R1		
Contact Block*			CR104PXC01F		
Name Plate*			CR104PXN2RP009 "Emergency Off"		
SS Wall Plate*			CR2940BD201D		
Selector Switch		-	P9SSMD0V		
Pilot Lamp (Green)		-	080BA9S6LV		
Battery	12 DC	-	BP7-12, GE Ind. Sys. 4001N43P001		2276547-14
Charger	12 DC	0.8	11027, GE Ind. Sys. 4001N44P001		2276547-16
* = Component parts of additional remote emergency OFF pushbuttons.					

All of the above parts are in stock at GE Supply Milwaukee (414)-527-6600.

5-2 Replacement Parts

See Table 5-2 for replacement parts for the Twin System main disconnect panel.

Table 5-2
LIST FOR REPLACEMENT PARTS

Part	Manufacturer Part Number		
Power Distribution Block	4000N51P062		
Neutral Block	4000N51P068		
CB1 - Circuit Breaker	SEHA36AT0150		
CB1 - 150 AMP Rating Plug	SRPE150A150		
CB1 – Optional 480V ONLY Rating Plug	SRPE150A125		
CB1 – Lug Qty (4)	TCAL18		
CB1 - Lug Qty (2)	TCAL18LV		
CB2 – 100A Circuit Breaker	FCV36TE100RT	+(6) FCALK13	
CB2 – CB3 20 AMP Circuit Breaker	FCV326TE020R2		
CB3 – 30 AMP Circuit Breaker	FCV326TE030	+(6) FCALK13	
CB6 – 15 AMP Rating Plug	FCV326TE015		
CB7 - V-Line Circuit Breaker	V07115		
CB8 - V-Line Circuit Breaker	V37110		
F1 and F4 Fuse Blocks			
F2, F10, and F11 Fuse Blocks	3007N36P001		
F3 Fuse Block			
F5 Fuse Block			
C1	CK95BE311J		
C2	CK09BE311J		
C3	CL04A310MJ		
CPT1	9T58E0506		
CPT2	9T58E0507		
CR1	4001N42P001		
T1	RL4RA040TJ	BTLF30C	
UPS and MUX BOX-Receptacle, Box, and Cover	3009N13P002	3009N14P001	3009N14P003
MAG MON and Modem-Receptacle, Box, and Cover	3009N13P001	3009N14P001	3009N14P003
Charger-Receptacle, Box, and Cover	3009N13P003	3009N14P001	3009N14P004
UPS Return-Receptacle, Box, and Cover	3009N12P001	3009N14P002	
Emergency "OFF" PB-On Cover	P9SPNRG		
Main Power and PDU Power "ON" PB's	P9SPNVG		
PDU, SCC, Cryo Cooler and Modem, UPS "ON-OFF" Selector Switches	P9SSMD0V		
PDU, SCC, Cryo Cooler and UPS, MUX BOX, MAG MON, MOD Light Operators, and Power Supply MOD's	P9SLVD	P9PTNVJLV	
Cord strain relief devices	Heyco 1237, 1517, 1140, 1154		

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	11-7-2003	L. R. H.	Manual Released
1	2-6-2004	L.R.H	Revised wiring diagram and layout drawing
2	3-27-2005	L.R.H	Changed 5A fuse GEHC FRU #
3	3-12-2006	L.R.H	Changed weight to 350 lbs.

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